

Sub-Thermocline Acoustic Backscatter Classification Using Lightweight Convolutional Models

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Abstract

We present a lightweight convolutional model for classifying sub-thermocline acoustic backscatter into 7 target classes. Trained on 12,400 labelled profiles collected in the Salish Sea between March 2023 to November 2024, the model achieves 94.2 per cent classification accuracy against a 87.6 per cent baseline, with a macro-averaged F1 score of 0.931. At 1.8 million parameters the model runs on-device in 40 milliseconds, making real-time classification feasible on battery-powered instrumentation.

1. Introduction

Acoustic backscatter profiling is widely used to characterise particulate and biological scatterers in the water column. Classification has conventionally been performed ashore, after recovery, because the models involved have been too large to run on the instrument itself.

Moving classification onto the instrument changes what is possible operationally: an instrument that can recognise a class of interest in real time can adapt its own sampling strategy, and can report summaries over low-bandwidth links rather than storing everything for later recovery.

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2. Related work

Prior approaches fall broadly into two groups: hand-engineered feature pipelines with classical classifiers, and larger convolutional models trained end to end. The former are efficient but brittle across sites; the latter generalise better but have not been deployable on instrument-class hardware.

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3. Dataset

Our dataset comprises 12,400 labelled profiles collected in the Salish Sea between March 2023 to November 2024. Profiles were recorded across a range of sea states and times of day, and were labelled by two annotators independently with disagreements resolved by a third.

We use a 70 / 15 / 15 train, validation, and test split, partitioned by deployment rather than by profile so that profiles from a single deployment cannot appear in more than one split.

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4. Method

The model is a one-dimensional convolutional network with 1.8 million parameters, operating directly on the range-binned backscatter series. Depthwise separable convolutions are used throughout to reduce parameter count.

Training used the Adam optimiser with a cosine learning-rate schedule over 60 epochs, a batch size of 128, and standard augmentation comprising range jitter, amplitude scaling, and additive noise.

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5. Results

Model	Accuracy	F1	Params
Feature pipeline (baseline)	87.6%	0.842	—
ResNet-18 (reference)	95.1%	0.940	11.7M
Ours	94.2%	0.931	1.8M

Our model reaches 94.2 per cent accuracy, within one percentage point of a reference ResNet-18 at approximately one sixth of the parameter count. Inference latency on the instrument's embedded processor is 40 milliseconds per profile.

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6. Discussion

The remaining error is concentrated in two visually similar classes, which together account for over half of all misclassifications. We expect that additional labelled data from outside the collection region would improve generalisation more than further architectural changes.

A limitation of this work is that all data originates from a single region. Performance in acoustically different environments remains to be established.

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7. Conclusion

We have shown that on-device classification of acoustic backscatter is practical on battery-powered instrumentation, at accuracy close to that of substantially larger models.

References

- [1] Vance, L. and Okonkwo, D. Range-binned representations for acoustic profiling. *Marine Sensing Letters*, 2023.
- [2] Mbeki, T. Annotation protocols for water column classification. *Proc. Ocean Instrumentation Workshop*, 2024.
- [3] Duplessis, S. Depthwise separable convolutions for embedded signal classification. *Embedded ML Review*, 2024.